Lesson 1 - Configuring the IDE

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Overview

Our goal is to create a unified development environment which supports thousands of microcontrollers and development boards. VS Code and all the extensions and support apps mentioned in this document are all free downloads.

Theory

Git / CVS software version control is necessary for software developers in 2026, but it's complicated and so we choose to present that lesson after we present foundational embedded development topics.

Prerequisites

Admin rights on the computer are needed to install some applications including Visual Studio. Extensions and Projects are stored in the user's profile, so admin is not necessary for extensions.

Objectives

- Install Microsoft Visual Studio Code.
- Install required extensions within VS Code.
- Clone the Microcontroller Training repository to the local hard drive.
- Familiarity with the local repository

Setting Up a New Computer

Installing VS Code

Note This section only needs to be done once for each computer.

1. Browse to `https://code.visualstudio.com/download`
2. Download VS Code for the OS the computer has installed. For Windows, use the one marked System instead of the one marked User and instead of the Windows Store version. It will be installed for all users and give the ability to edit important files.
3. **Right-click the downloaded file** and choose **Run As Administrator**.
4. Click **Yes** to give the popup window permission.
5. Click **Accept** to agree to the license agreement.
6. Leave the default location to install VS Code and click **Next**.
7. Click the Create a Desktop Icon and "Open with Code" options then click **Next** and then click **Install**.
8. Leave Launch Visual Studio Code checked and then click **Finish**.

Installing Extensions

Note This section only needs to be done once for each computer.

1. Open VS Code.
2. From the left-hand toolbar, click the Extensions icon.

Figure 1: PlatformIO Extensions Icon

3. Search Platform IO and install the Platform IO extension

4. Trust Platform IO if a popup box appears.

5. Click Reload Extensions after Platform IO installs.

6. From the left-hand toolbar, click the Extensions icon (See Figure 1).

7. Search for the Microsoft C/C++ Extension and install it if it's not already installed. It's Microsoft, so no trust needed.

8. Search for the GitHub Copilot Chat extension and install it.

9. The VS Code installed extensions should look like Figure 2 when you're done.

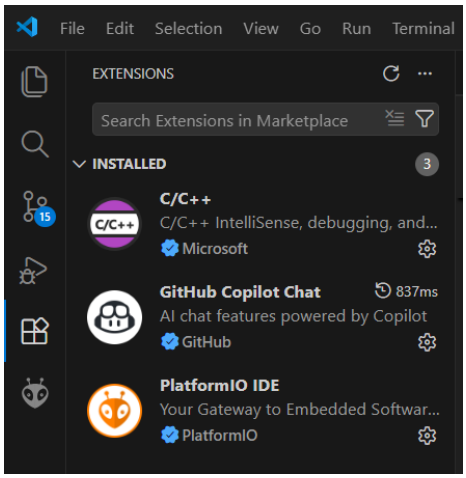

Figure 2: Installed Extensions

10. Restart VS Code.

11. Click on the Platform IO Icon (figure 3) to initialize the Platform IO core.

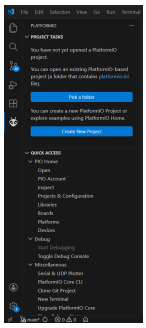

Figure 3: PlatformIO Extensions

Cloning the Microcontroller Training Repository

A repository is the collection of code and other files associated with a project. "Cloning the Microcontroller Training Repository" means literally copying the files in this location to our local hard drive.

We have chosen the location `C:\Users\userID/Documents/PlatformIO/Projects` to be the location where we will clone the repository of lessons because that's where Platform IO puts them by default.

1. Create the folder `C:/Users/userID/Documents/PlatformIO/Projects` - replace **userID** with the name of the userID that you logged in with.

2. Browse to this location: `https://github.com/CrazyUncleBurton/Microcontroller-Training`

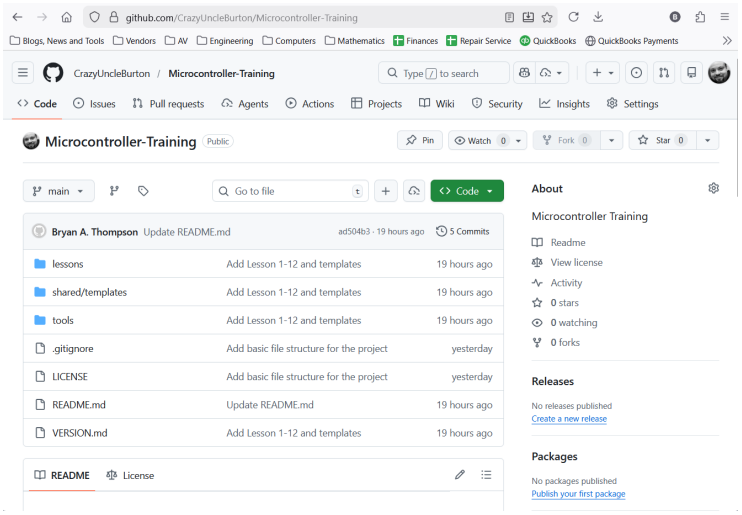

Figure 4: The Microcontroller GitHub Repository

3. Click the green Code button.

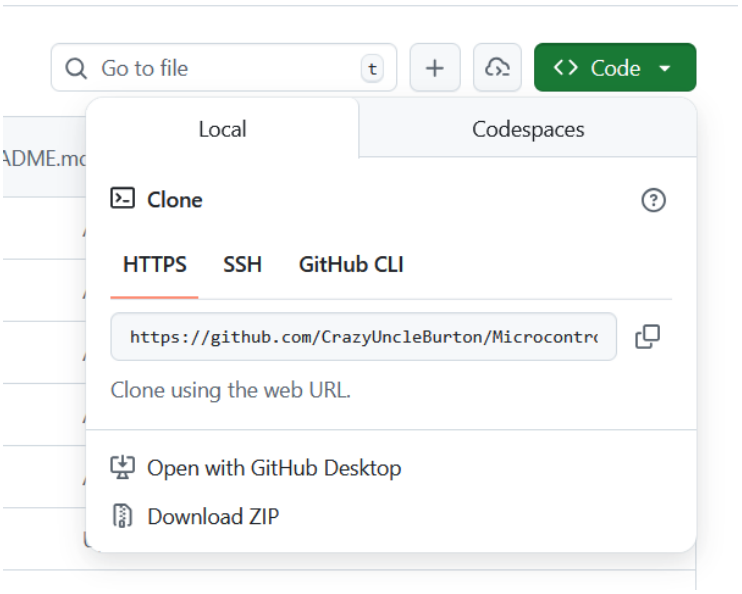

Figure 5: Link to Download Zip

4. Click **Download Zip**. Remember the location where you save the file.

5. Extract the contents of the Zip and place the **Microcontroller Training** folder in this folder: `C:/Users/CuserID/Documents/PlatformIO/Projects` and replace **userID** with the userID that you logged in with.

6. Go to the folder `C:/Users/userID/Documents/PlatformIO/Projects` - replace **userID** with the userID that you logged in with. It should look like this:

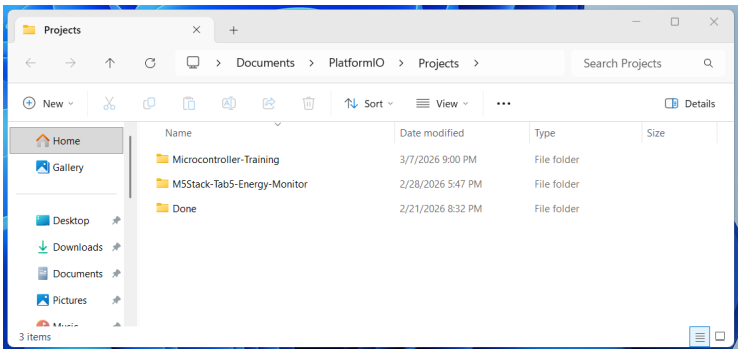

Figure 6: Link to the Microcontroller Training Project Folder

7. The lessons, including this one, are located in the **lessons** folder:

8. Browse into the lesson **02-hello-world**.

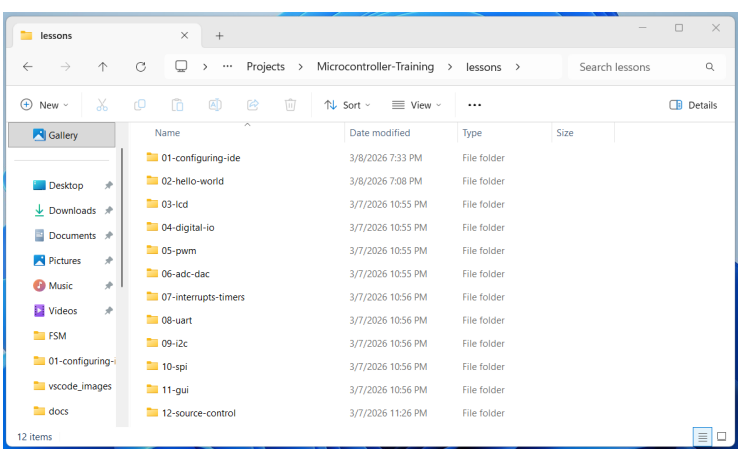

Figure 7: The Lessons folder

9. Look at the file structure includes the Word Doc / PDF of the lesson at the top level. The README.md file contains lesson-specific notes. The assets folder contains any files we might need, like schematics, datasheets, etc. The "code" folder contains the code we will compile. There will likely be several programs for each lesson.

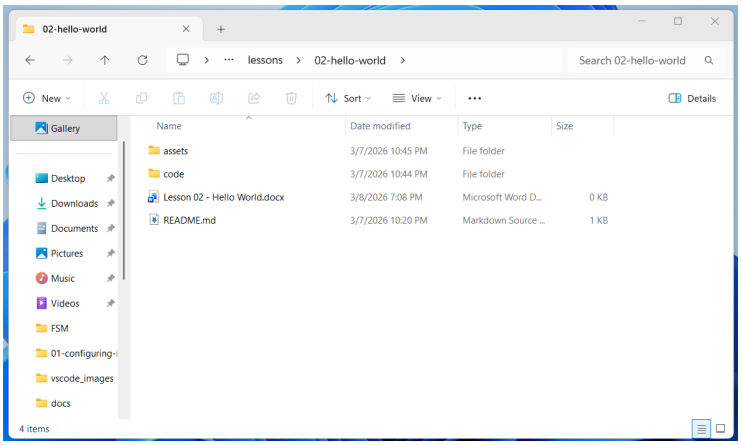

Figure 8: Lesson 02 - Hello World

Conclusion

This completes the software installation process. We will start the first program in Lesson 02 - Getting Started.